NYC Taxi Fare Prediction Using Machine Learning Models

A Lab Project Report submitted for the partial fulfillment of the degree of

Bachelor of Technology

in

Computer Science and Engineering

Submitted by

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CERTIFICATE

This is to certify that the Lab Project entitled **"NYC Taxi Fare Prediction Using Machine Learning Models"** submitted by **S.NIRUPAMA (S221001)** and **K.BHAVYA (S221081)** in partial fulfillment of the requirements for the Data Science with Python Lab Project.

Faculty Signature

ABSTRACT

This project aims to predict the total cost (total amount) of taxi trips using machine learning techniques. The dataset consists of various trip-related features such as passenger count, trip distance, pickup and drop-off details, and fare components.

Data preprocessing was performed to clean and prepare the dataset. Missing values were handled, irrelevant columns were removed, and categorical variables were encoded. Additionally, feature extraction was carried out by deriving new features from existing columns, such as time and date-based attributes, to improve model performance.

Data visualization techniques were used to analyze patterns and relationships between input features and the target variable.

Multiple regression models, including Linear Regression, Multiple Linear Regression, and Polynomial Regression, were implemented to predict the total amount. The dataset was split into training and testing sets, and model performance was evaluated using the R^2 score.

Among the models, Multiple Linear Regression achieved the best performance with an R^2 score of 0.9484550, indicating a strong relationship between the selected features and the target variable. The prediction of total cost is primarily based on features such as passenger count, trip distance, and other relevant trip attributes.

This project demonstrates how machine learning models can effectively predict taxi trip costs and highlights the importance of feature engineering in improving model accuracy.

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Chapter 1

Introduction

This chapter introduces the project background and objectives.

1.1 Background

The rapid growth of urban transportation systems has led to an increased demand for efficient and cost-effective taxi services. Accurately predicting the cost of a taxi trip is an important problem for both service providers and passengers, as it helps in better planning, transparency, and decision-making.

With the availability of large-scale transportation datasets such as taxi trip records, it is possible to analyze patterns and build predictive models. These datasets contain valuable information including passenger count, trip distance, pickup and drop-off details, and fare components, which influence the total cost of a trip.

Machine learning techniques, particularly regression models, play a significant role in solving such prediction problems. By learning the relationship between input features and the target variable, these models can estimate the total trip cost with high accuracy.

In this project, regression techniques such as Linear Regression, Multiple Linear Regression, and Polynomial Regression are used to model the relationship between trip features and the total amount. The use of feature extraction further enhances the model's ability to capture patterns in the data.

This background highlights the importance of data-driven approaches in transportation systems and sets the foundation for predicting taxi trip costs effectively.

1.2 Problem Statement

The objective of this project is to develop a machine learning model to predict the total cost (total amount) of taxi trips based on various trip-related features such as passenger count, trip distance, and other relevant attributes.

Accurate prediction of taxi fares is challenging due to the influence of multiple factors and potential non-linear relationships between variables. Therefore, the problem is to analyze the given dataset, extract meaningful features, and apply suitable regression techniques to build a model that can predict the total trip cost with high accuracy.

The performance of the model is evaluated using the R^2 score, with the goal of achieving a high prediction accuracy.

1.3 Objectives

- To analyze the taxi trip dataset and understand the key features affecting the total cost.
- To perform data preprocessing, including handling missing values, encoding categorical variables, and cleaning the dataset.
- To perform feature extraction by creating new features from existing data to improve model performance.
- To visualize the dataset and identify patterns and relationships between input features and the target variable.
- To build and implement regression models such as Linear Regression, Multiple Linear Regression, and Polynomial Regression.
- To evaluate model performance using the R^2 score.
- To identify the best-performing model for predicting the total taxi trip cost.

1.4 Organisation of Report

This report is organized into several sections to clearly present the workflow of the project.

Chapter 1 introduces the background, problem statement, and objectives of the project.

Chapter 2 describes the data preprocessing steps, including data cleaning, handling missing values, encoding categorical variables, and feature extraction.

Chapter 3 presents data visualization techniques used to understand patterns and relationships in the dataset.

Chapter 4 explains the model fitting process, where regression models such as Linear Regression, Multiple Linear Regression, and Polynomial Regression are applied and evaluated using the R^2 score.

Finally, Chapter 5 provides the references used in this project.

Chapter 2

Data Preprocessing

Data preprocessing is a crucial step in any data science project. In this NYC TLC taxi analysis and prediction project, the raw dataset was cleaned and transformed into a structured format suitable for machine learning.

- **Using Raw Dataset:** The project begins with an uncleaned NYC TLC dataset containing trip-related information such as passenger count, trip distance, timestamps, and fare details.
- **Handling Missing Values:** Missing values in numerical columns were identified and replaced using median imputation to maintain robustness and reduce the effect of outliers.
- **Removing Duplicates:** Duplicate records were detected and removed to ensure data consistency and avoid redundancy in the dataset.
- **Identifying Negative Values:** Invalid entries such as negative values in features like trip distance and fare amount were identified and filtered out.
- **Fixing Date and Time Format:** Datetime columns were properly converted into standard datetime format to enable time-based analysis and feature extraction.
- **Feature Engineering:** New features were created from existing columns, such as extracting hour, day, and weekday from datetime data to capture temporal patterns.
- **Handling Outliers:** Outliers were capped based on percentile thresholds to reduce their impact on model performance.
- **Encoding Categorical Variables:** Categorical columns such as store_and_fwd_flag were converted into numerical format using encoding techniques.
- **Final Data Validation:** After preprocessing, the dataset was rechecked to ensure no missing values, inconsistencies, or invalid entries remained.

These preprocessing steps ensured that the dataset was clean, reliable, and ready for visualization and model building.

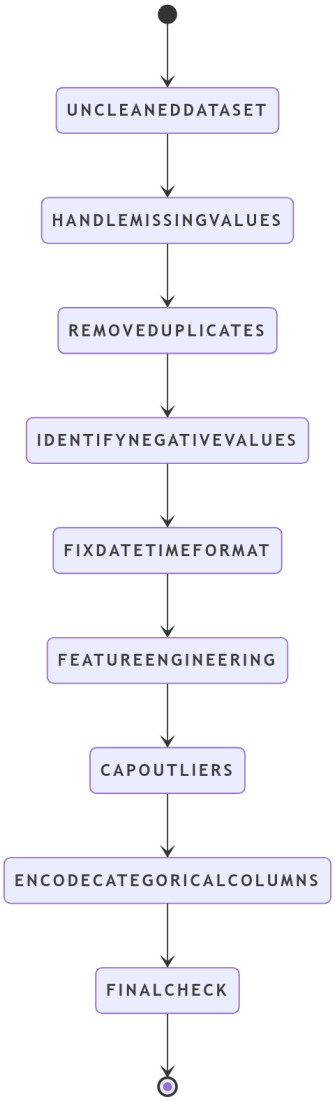

Figure 2.1: preprocessing workflow

Chapter 3

Data Visualization

Data visualization is an important step to understand patterns, relationships, and trends in the dataset. In this project, various visualization techniques were used to analyze the NYC TLC taxi dataset.

- **Using Preprocessed Dataset:** The cleaned dataset obtained after preprocessing was used for visualization to ensure accurate and meaningful analysis.
- **Univariate Analysis:** Individual features such as trip distance, passenger count, and total amount were analyzed using histograms to understand their distribution.
- **Bivariate Analysis:** Relationships between two variables were examined using scatter plots, such as trip distance vs total_amount, to observe how cost varies with distance.
- **Multivariate Analysis:** Correlation between multiple features was analyzed using heatmaps to identify strong and weak relationships among variables.
- **Trend Analysis:** Time-based patterns were studied by analyzing features derived from datetime columns, such as peak hours and daily trends.
- **Insight Extraction:** Visualization helped identify that total cost increases with trip distance and is influenced by multiple factors.
- **Outlier Observation:** Plots such as boxplots were used to detect extreme values and understand their impact on the dataset.

These visualizations provided valuable insights that supported feature selection and improved model building.

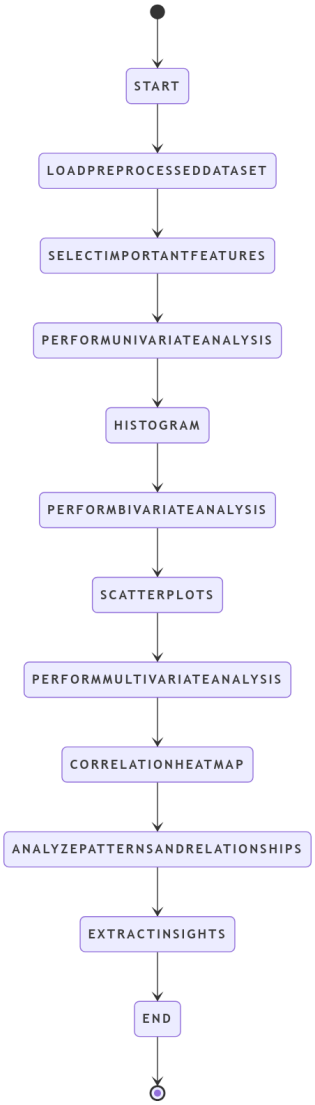

Figure 3.1: Data Insights Workflow

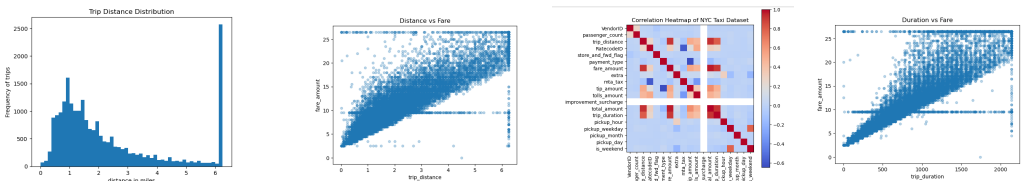

Figure 3.2: Visualization Graphs

Chapter 4

Model Fitting

Model fitting involves training machine learning models on the prepared dataset to predict the target variable. In this project, regression techniques were used to predict the total taxi trip cost (total_amount).

- **Feature Selection:** Relevant features such as passenger count, trip distance, and other trip-related attributes were selected as input variables, while total_amount was used as the target variable.
- **Data Splitting:** The dataset was divided into training and testing sets using an 80:20 ratio to evaluate model performance on unseen data.
- **Linear Regression:** A basic regression model was applied to understand the linear relationship between input features and the target variable.
- **Multiple Linear Regression:** This model used multiple input features simultaneously to improve prediction accuracy and capture more complex relationships.
- **Polynomial Regression:** Polynomial features were generated to capture non-linear relationships between variables, enhancing model performance.
- **Model Prediction:** Each model was trained on the training dataset and used to predict values on the testing dataset.
- **Evaluation Metric:** Model performance was evaluated using the R^2 score, which measures how well the model explains the variance in the target variable.
- **Model Comparison:** The performance of all models was compared based on their R^2 scores.
- **Best Model Selection:** Multiple Linear Regression achieved the highest R^2 score of 0.9484550, making it the best-performing model for this project.

These steps ensured accurate prediction of taxi trip cost and demonstrated the effectiveness of regression techniques.

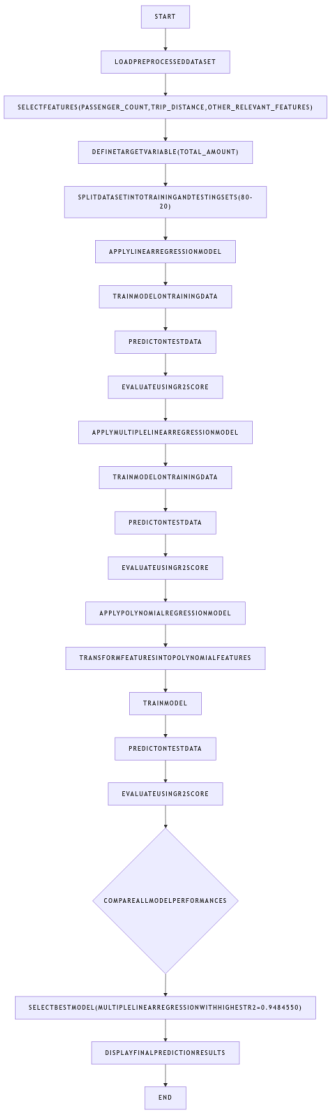

Figure 4.1: Work Flow in Model Fitting

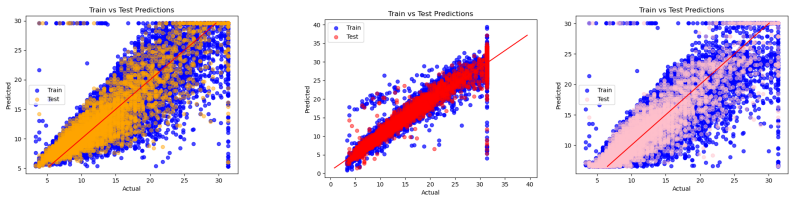

Figure 4.2: ModelFitting Graphs (Linear , MultiLinear , PolynomialReg)

Chapter 5

References

5.1 ipy notebooks

- `http://localhost:8890/notebooks/cse2c_NYCTLC_Preprocessing.ipynb`
- http://localhost:8889/notebooks/cse2c_NYCTLC_visualizations1.ipynb
- http://localhost:8888/notebooks/CSE2C_NYC_TLC_ModelFitting.ipynb

5.2 Dataset

- https://share.google/INmI9scejyn55vVyF

5.3 Github Repositary

- https://github.com/kandregula-bhavya/NYC-Taxi-Cost-Prediction-ML.git
- https://www.analyticsvidhya.com/blog/2021/10/everything-you-need-to-know-about-linear-regression/